

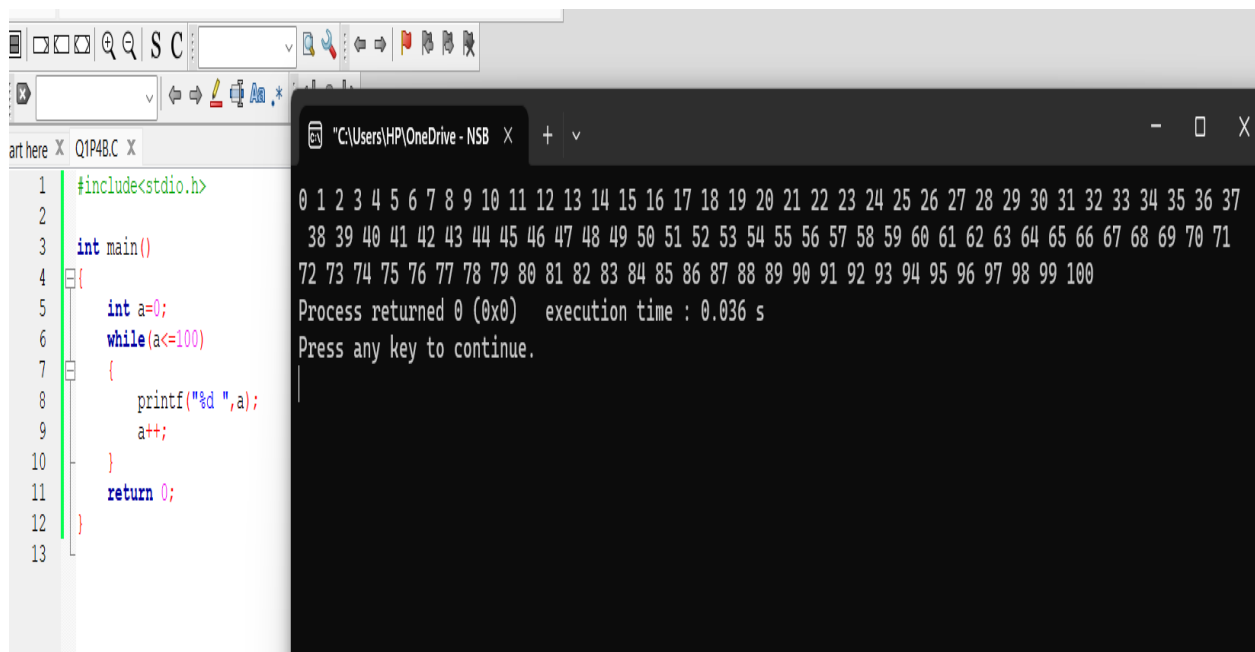
# PRACTICAL 5

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## Section A

(Q1) - While Loop

```
int main()
{
    int a=0;
    while(a<=100)
    {
        printf("%d ",a);
        a++;
    }
    return 0;
}
```



The screenshot shows a C program being executed in a terminal window. The program uses a while loop to print numbers from 0 to 100. The output is displayed in a single line, with numbers separated by spaces. The terminal window title is "C:\Users\HP\OneDrive - NSB". The program code is visible in the background, showing the while loop and the printf statement.

```
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37
38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71
72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
Process returned 0 (0x0)   execution time : 0.036 s
Press any key to continue.
```

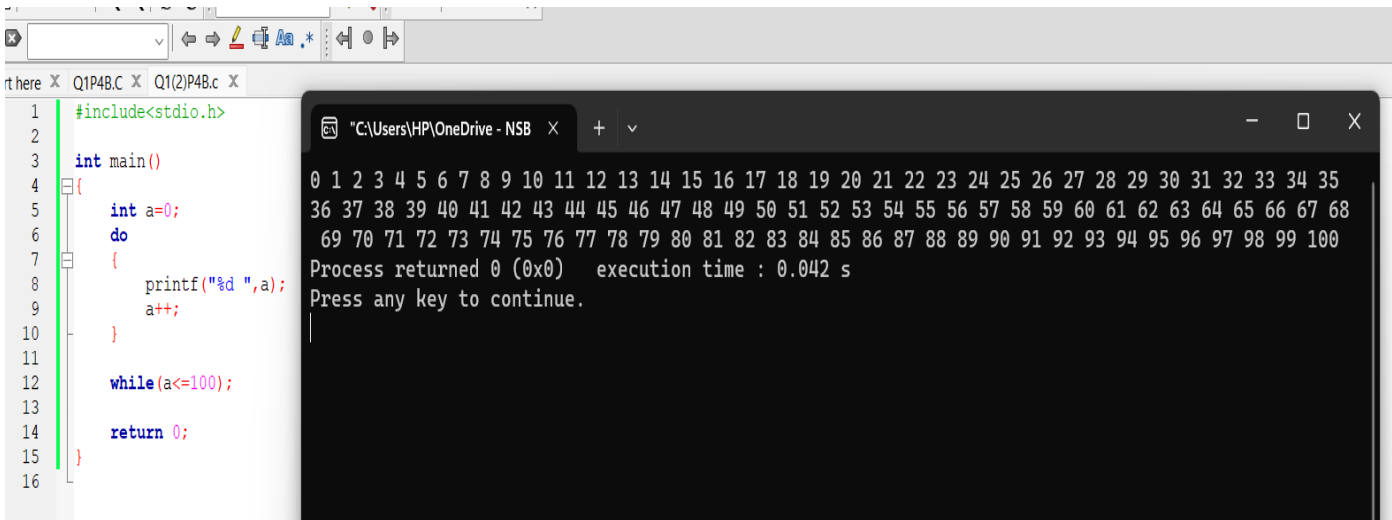
- Do while

```
#include<stdio.h>

int main()
{
    int a=0;
    do
    {
        printf("%d ",a);
        a++;
    }

    while(a<=100);

    return 0;
}
```



The screenshot shows a C program being executed in a Windows environment. On the left, a code editor displays the source code for a program that prints numbers from 0 to 100 using a do-while loop. The code is as follows:

```
1 #include<stdio.h>
2
3 int main()
4 {
5     int a=0;
6     do
7     {
8         printf("%d ",a);
9         a++;
10    }
11
12    while(a<=100);
13
14    return 0;
15 }
16
```

On the right, a command prompt window shows the output of the program. It displays the numbers 0 through 100, followed by the message "Process returned 0 (0x0) execution time : 0.042 s" and "Press any key to continue." The numbers are arranged in three rows: the first row contains 0-35, the second row contains 36-68, and the third row contains 69-100.

- **For Loop**

```
#include<stdio.h>
```

```
int main()
{
    int a;
    for(a=0;a<=100;a++)
    {
        printf("%d ",a);

    }
    return 0;
}
```

The screenshot shows a code editor with a C program and its output in a terminal window. The code defines a `main` function that uses a `for` loop to print integers from 0 to 100, separated by spaces. The terminal output displays the sequence of numbers from 0 to 100, followed by the message "Process returned 0 (0x0) execution time : 0.045 s" and "Press any key to continue.".

```
1
2
3
4 #include<stdio.h>
5
6 int main()
7 {
8     int a;
9     for(a=0;a<=100;a++)
10    {
11        printf("%d ",a);
12
13    }
14    return 0;
15 }
16
```

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39  
40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76  
77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100  
Process returned 0 (0x0) execution time : 0.045 s  
Press any key to continue.

```

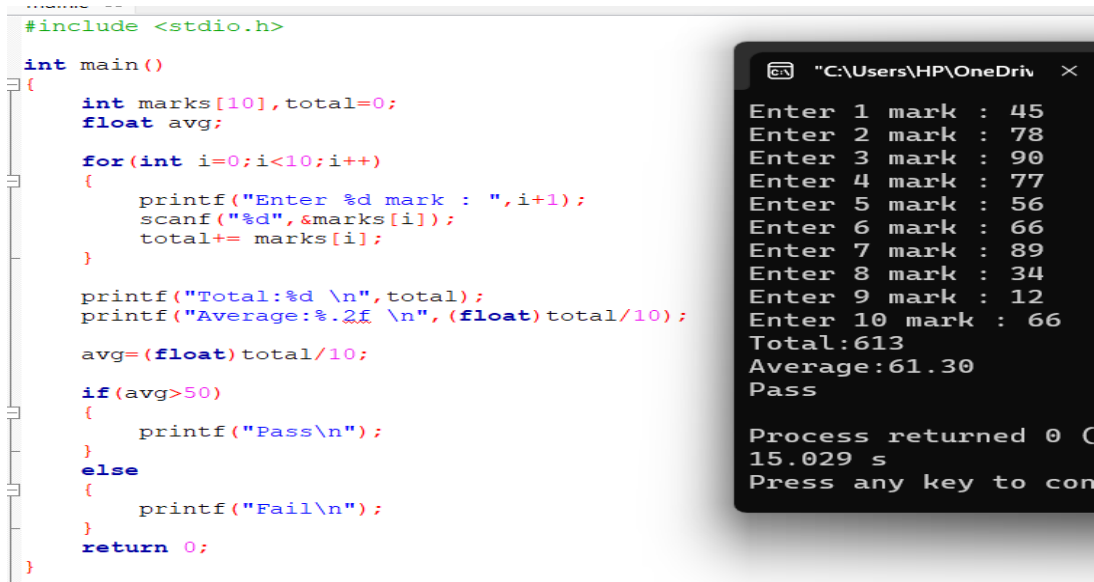
Q2) #include <stdio.h>
int main()
{
    int marks[10],total=0;
    float avg;

    for(int i=0;i<10;i++)
    {
        printf("Enter %d mark : ",i+1);
        scanf("%d",&marks[i]);
        total+= marks[i];
    }
    printf("Total:%d \n",total);
    printf("Average:%.2f \n",(float)total/10);

    avg=(float)total/10;

    if(avg>50)
    {
        printf("Pass\n");
    }
    else
    {
        printf("Fail\n");
    }
    return 0;
}

```



The image shows a screenshot of a C program being executed. On the left, the source code is displayed in a text editor with syntax highlighting. On the right, a terminal window shows the program's output, including prompts for marks, the calculated total and average, and the final pass/fail result.

```

#include <stdio.h>

int main()
{
    int marks[10],total=0;
    float avg;

    for(int i=0;i<10;i++)
    {
        printf("Enter %d mark : ",i+1);
        scanf("%d",&marks[i]);
        total+= marks[i];
    }

    printf("Total:%d \n",total);
    printf("Average:%.2f \n", (float)total/10);

    avg=(float)total/10;

    if(avg>50)
    {
        printf("Pass\n");
    }
    else
    {
        printf("Fail\n");
    }
    return 0;
}

```

Terminal Output:

```

Enter 1 mark : 45
Enter 2 mark : 78
Enter 3 mark : 90
Enter 4 mark : 77
Enter 5 mark : 56
Enter 6 mark : 66
Enter 7 mark : 89
Enter 8 mark : 34
Enter 9 mark : 12
Enter 10 mark : 66
Total:613
Average:61.30
Pass

Process returned 0 (
15.029 s
Press any key to con

```

```

Q3) #include <stdio.h>
    int main()
    {
        int number,i;
        int factorial=1;

        printf("Enter a number: ");
        scanf("%d",&number);
        if(number<0)
        {
            printf("Not calculatable\n");
        }
        else if(number==0)
        {
            printf("Factorial of %d is %d \n",number,1);
        }
        else
        {
            for(i=1;i<=number;i++)
            {
                factorial *= i;
            }
            printf("Factorial of %d is %d \n",number,factorial);
        }
        return 0;
    }

```

The screenshot shows a C program being executed in a terminal window. The program prompts the user to enter a number, and the user has entered 5. The program then calculates the factorial of 5, which is 120, and displays the result. The terminal window title is "C:\Users\HP\OneDrive - NSB".

```

C X main.c X main.c X
1 | #include <stdio.h>
2 |
3 | int main()
4 | {
5 |     int number,i;
6 |     int factorial=1;
7 |
8 |     printf("Enter a number: ");
9 |     scanf("%d",&number);
10 |
11 |     if(number<0)
12 |     {
13 |         printf("Not calculatable\n");
14 |     }
15 |     else if(number==0)
16 |     {
17 |         printf("Factorial of %d is %d \n",number,1);
18 |     }
19 |     else
20 |     {
21 |         for(i=1;i<=number;i++)
22 |         {
23 |             factorial *= i;
24 |         }
25 |         printf("Factorial of %d is %d \n",number,factorial);
26 |     }
27 |     return 0;
28 | }

```

Terminal Output:

```

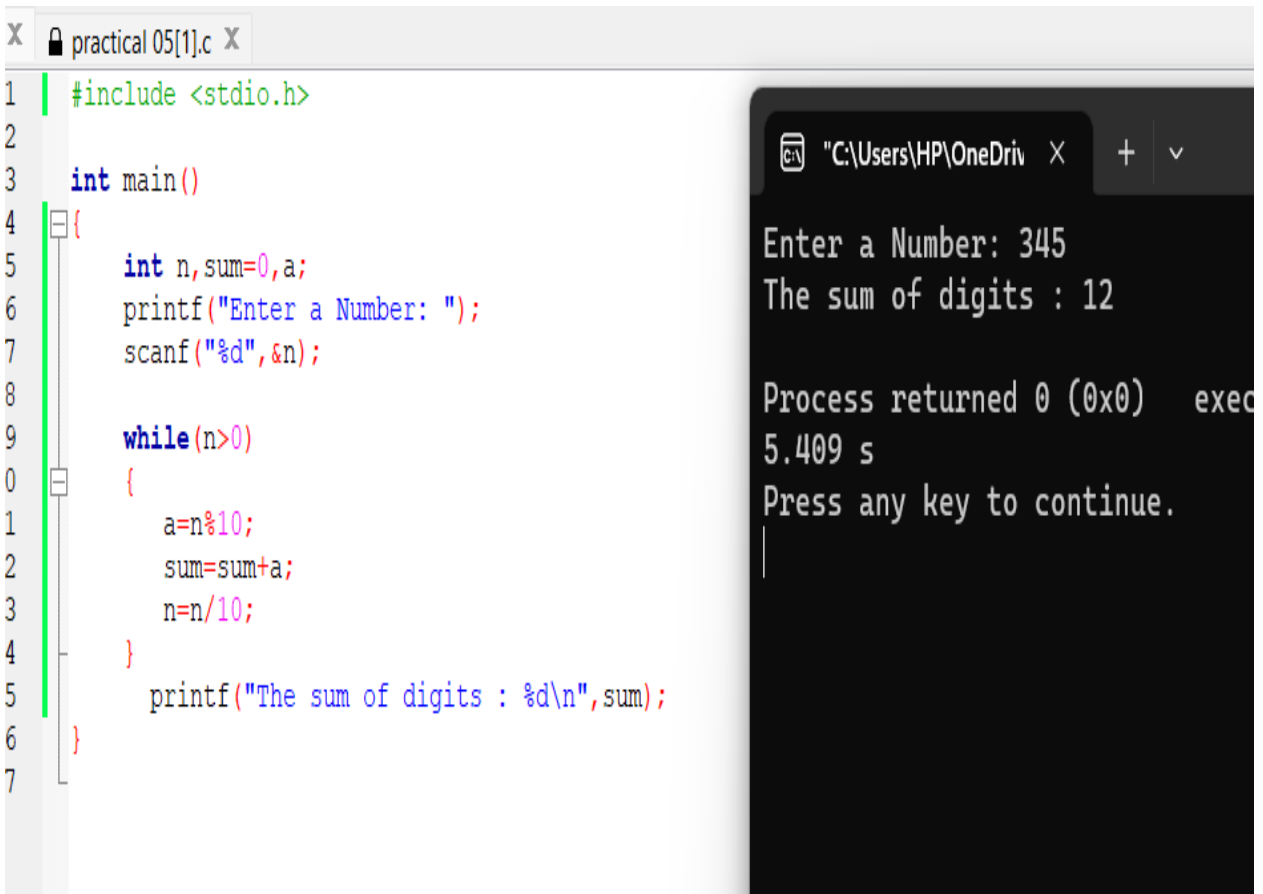
"C:\Users\HP\OneDrive - NSB
Enter a number: 5
Factorial of 5 is 120

Process returned 0 (0x0)
Press any key to continue

```

```
Q4) #include <stdio.h>
int main()
{
    int n,sum=0,a;
    printf("Enter a Number: ");
    scanf("%d",&n);

    while(n>0)
    {
        a=n%10;
        sum=sum+a;
        n=n/10;
    }
    printf("The sum of digits : %d\n",sum);
}
```



The image shows a screenshot of a C program being executed. On the left, a code editor window titled "practical 05[1].c" displays the source code. The code is a C program that calculates the sum of the digits of a number. It includes the standard input/output header, defines a main function, and uses a while loop to extract digits from the number 345. On the right, a terminal window shows the program's output. It prompts the user to "Enter a Number:", which is 345. It then displays "The sum of digits : 12". Below this, it shows the process return status as 0 (0x0) and the execution time as 5.409 seconds. The terminal also prompts the user to "Press any key to continue.".

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n,sum=0,a;
6      printf("Enter a Number: ");
7      scanf("%d",&n);
8
9      while(n>0)
10     {
11         a=n%10;
12         sum=sum+a;
13         n=n/10;
14     }
15     printf("The sum of digits : %d\n",sum);
16 }

Enter a Number: 345
The sum of digits : 12

Process returned 0 (0x0)   exec
5.409 s
Press any key to continue.
```

Q5)

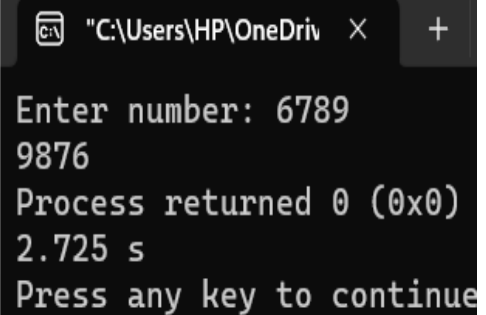
```
#include <stdio.h>
int main()
{
    int n,reverse;
    printf("Enter number: ");
    scanf("%d",&n);
    do
    {
        reverse=n%10;
        printf("%d",reverse);
        n=n/10;
    }

    while(n>0);
    return 0;
}
```

```
#include <stdio.h>

int main()
{
    int n,reverse;
    printf("Enter number: ");
    scanf("%d",&n);
    do
    {
        reverse=n%10;
        printf("%d",reverse);
        n=n/10;
    }

    while(n>0);
    return 0;
}
```



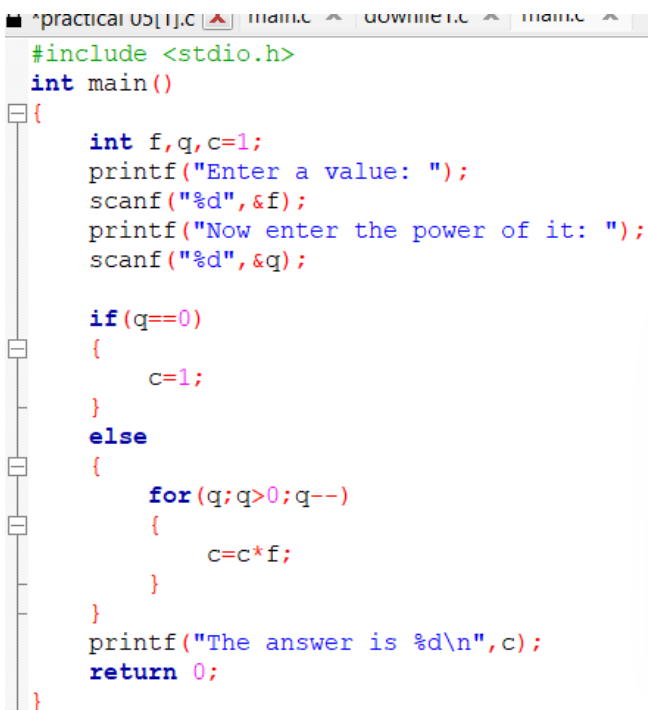
```
"C:\Users\HP\OneDrive" X +
Enter number: 6789
9876
Process returned 0 (0x0)
2.725 s
Press any key to continue
```

```

Q6) #include <stdio.h>
int main()
{
    int f,q,c=1;
    printf("Enter a value: ");
    scanf("%d",&f);
    printf("Now enter the power of it: ");
    scanf("%d",&q);

    if(q==0)
    {
        c=1;
    }
    else
    {
        for(q;q>0;q--)
        {
            c=c*f;
        }
    }
    printf("The answer is %d\n",c);
    return 0;
}

```

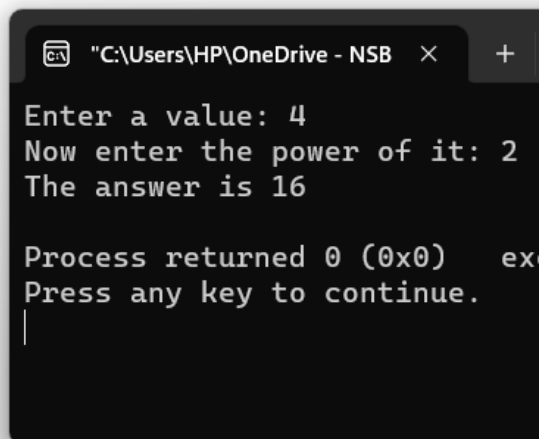


```

#include <stdio.h>
int main()
{
    int f,q,c=1;
    printf("Enter a value: ");
    scanf("%d",&f);
    printf("Now enter the power of it: ");
    scanf("%d",&q);

    if(q==0)
    {
        c=1;
    }
    else
    {
        for(q;q>0;q--)
        {
            c=c*f;
        }
    }
    printf("The answer is %d\n",c);
    return 0;
}

```



```

"C:\Users\HP\OneDrive - NSB"
Enter a value: 4
Now enter the power of it: 2
The answer is 16

Process returned 0 (0x0)   ex
Press any key to continue.

```



```

Q7) #include <stdio.h>
    int main()
    {
        int n1=0,n2=1,n3,count,num;

        printf("Enter the first 10 numbers of fibonanci \n");
        scanf("%d",&num);

        printf("\n%d\n%d\n",n1,n2);

        for(count=3;count<=num;count++)
        {
            n3=n1+n2;
            printf("%d\n",n3);

            n1 = n2;
            n2 = n3;
        }
        return 0;
    }

```

```

*practical 05[1].c x main.c x dowhile1.c x main.c x main.c x
#include <stdio.h>
int main()
{
    int n1=0,n2=1,n3,count,num;

    printf("Enter the first 10 numbers of fibonanci \n");
    scanf("%d",&num);

    printf("\n%d\n%d\n",n1,n2);

    for(count=3;count<=num;count++)
    {
        n3=n1+n2;
        printf("%d\n",n3);

        n1 = n2;
        n2 = n3;
    }
    return 0;
}

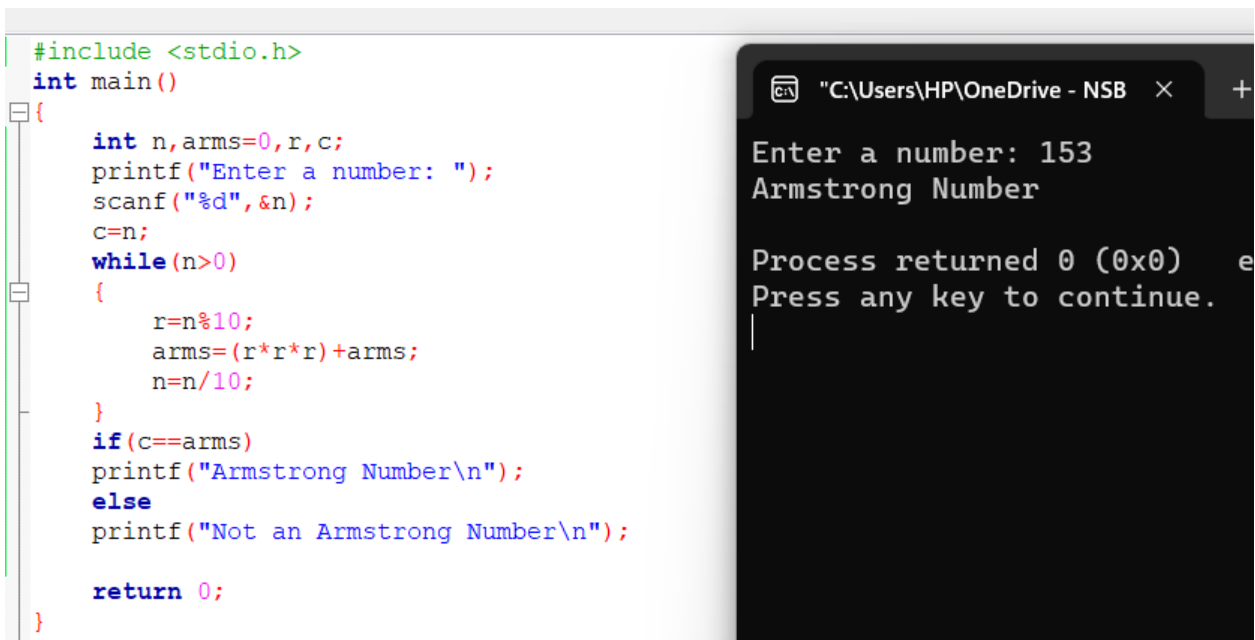
"C:\Users\HP\OneDriv x + v - □
Enter the first 10 numbers of fibonanci
10
0
1
1
2
3
5
8
13
21
34
55
89
144
Process returned 0 (0x0)   execution time :
3.311 s

```

Q8)

```
#include <stdio.h>
int main()
{
    int n,arms=0,r,c;
    printf("Enter a number: ");
    scanf("%d",&n);
    c=n;
    while(n>0)
    {
        r=n%10;
        arms=(r*r*r)+arms;
        n=n/10;
    }
    if(c==arms)
        printf("Armstrong Number\n");
    else
        printf("Not an Armstrong Number\n");

    return 0;
}
```



```
#include <stdio.h>
int main()
{
    int n,arms=0,r,c;
    printf("Enter a number: ");
    scanf("%d",&n);
    c=n;
    while(n>0)
    {
        r=n%10;
        arms=(r*r*r)+arms;
        n=n/10;
    }
    if(c==arms)
        printf("Armstrong Number\n");
    else
        printf("Not an Armstrong Number\n");

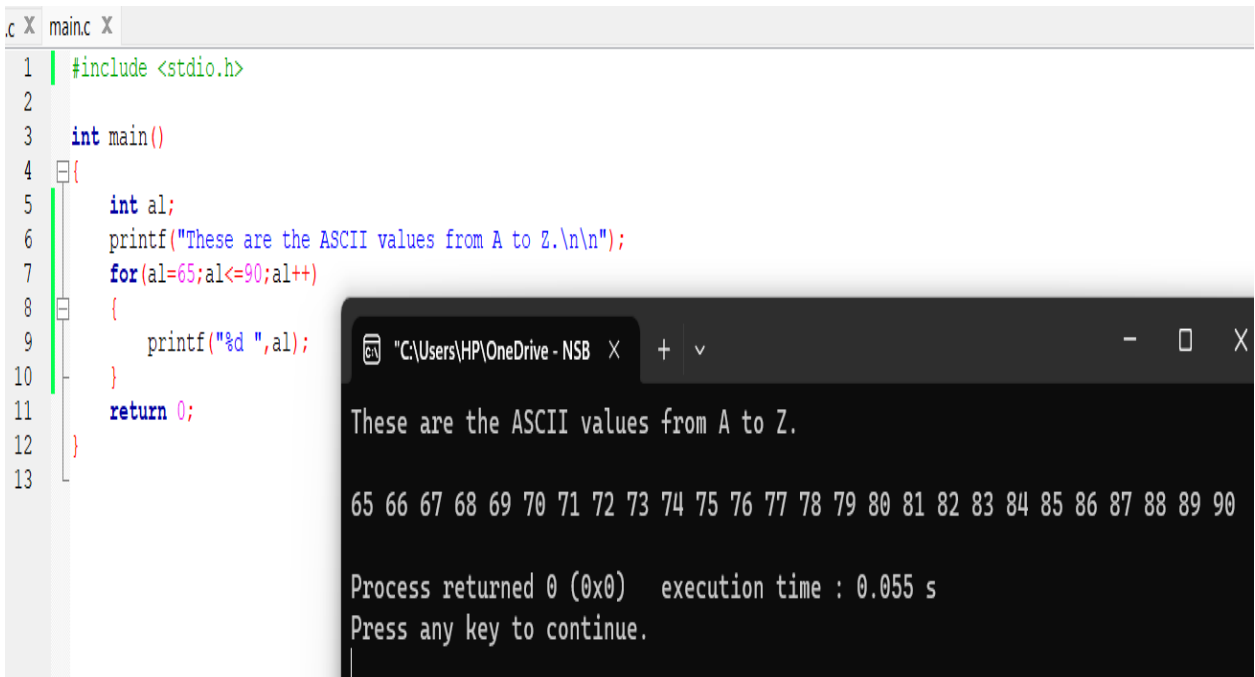
    return 0;
}
```

Enter a number: 153  
Armstrong Number  
Process returned 0 (0x0) Press any key to continue.

Q9)

```
#include <stdio.h>
```

```
int main()
{
    int al;
    printf("These are the ASCII values from A to Z.\n\n");
    for(al=65;al<=90;al++)
    {
        printf("%d ",al);
    }
    return 0;
}
```



The screenshot shows a code editor window with a file named 'main.c'. The code is a C program that prints the ASCII values of uppercase letters A through Z. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int al;
6     printf("These are the ASCII values from A to Z.\n\n");
7     for(al=65;al<=90;al++)
8     {
9         printf("%d ",al);
10    }
11    return 0;
12 }
13
```

Below the code editor, a terminal window is open, showing the output of the program. The output is:

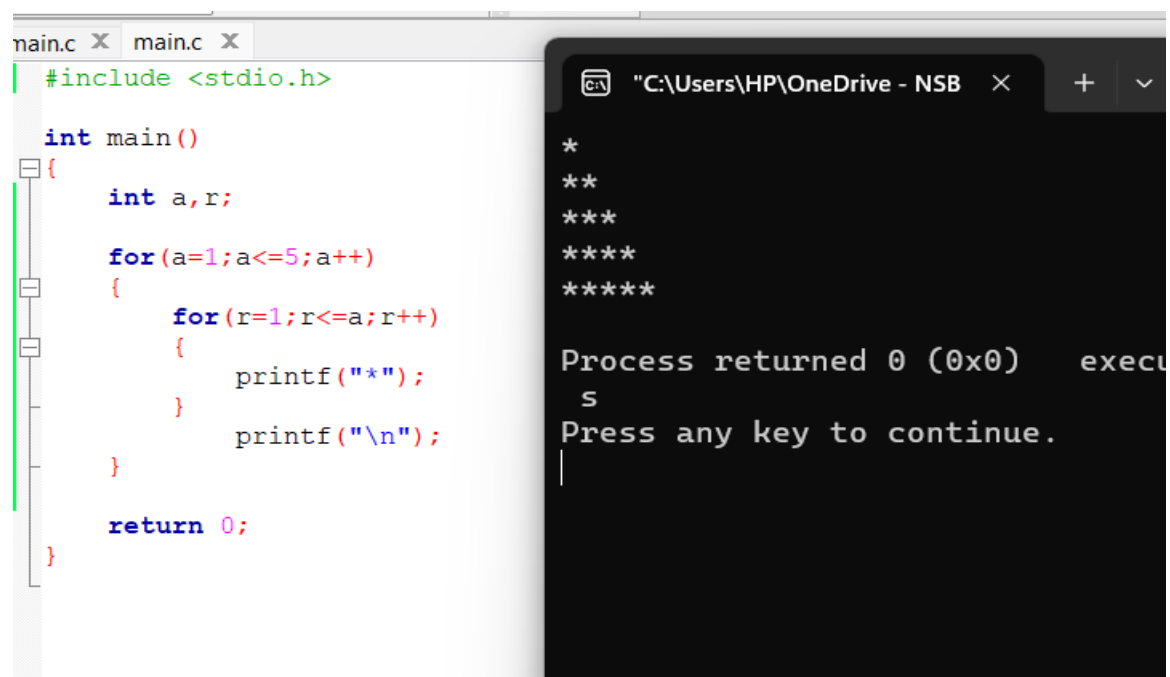
```
These are the ASCII values from A to Z.

65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90

Process returned 0 (0x0)   execution time : 0.055 s
Press any key to continue.
```

Q10)

```
#include <stdio.h>
int main()
{
    int a,r;
    for(a=1;a<=5;a++)
    {
        for(r=1;r<=a;r++)
        {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```



The screenshot shows a code editor with a C program and its output in a terminal window. The code is a nested loop that prints a pattern of asterisks. The terminal output shows the pattern: a single asterisk on the first line, two on the second, three on the third, four on the fourth, and five on the fifth. Below the pattern, the terminal displays the message "Process returned 0 (0x0) execution completed. Press any key to continue." and a cursor is visible on the line following the message.

```
main.c X main.c X
#include <stdio.h>

int main()
{
    int a,r;

    for(a=1;a<=5;a++)
    {
        for(r=1;r<=a;r++)
        {
            printf("*");
        }
        printf("\n");
    }

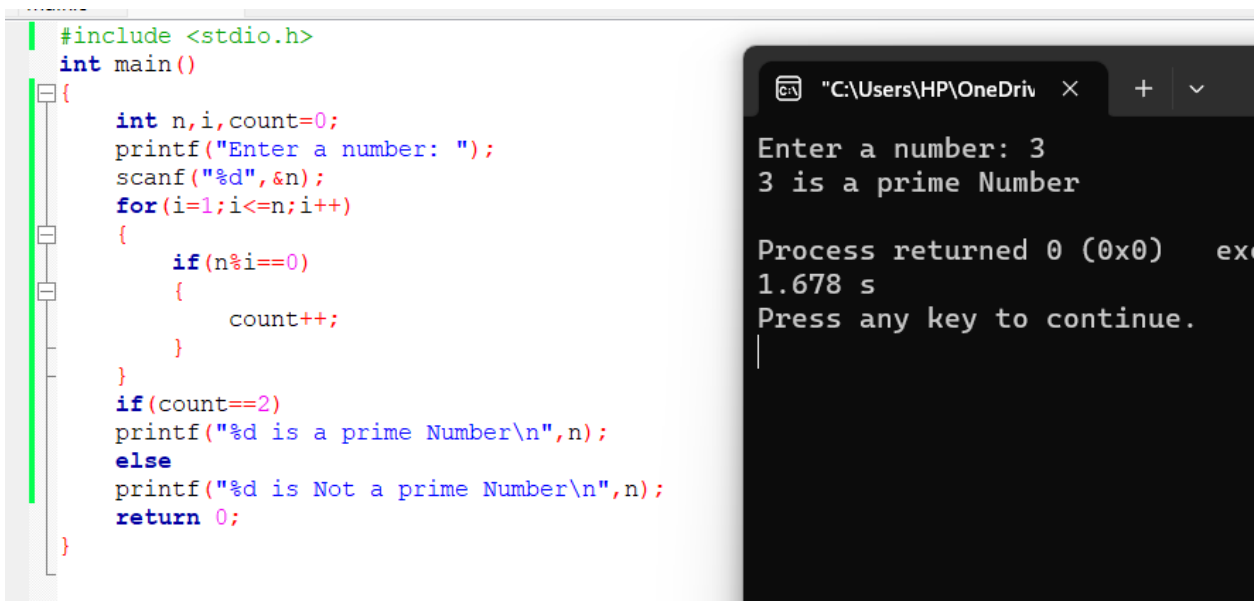
    return 0;
}

*
**
***
****
*****

Process returned 0 (0x0)   execution completed.
Press any key to continue.
|
```

Q11)

```
#include <stdio.h>
int main()
{
    int n,i,count=0;
    printf("Enter a number: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            count++;
        }
    }
    if(count==2)
    printf("%d is a prime Number\n",n);
    else
    printf("%d is Not a prime Number\n",n);
    return 0;
}
```



The image shows a screenshot of a C program and its execution. On the left, a code editor displays the source code for a prime number checker. The code includes `<stdio.h>`, defines `main()`, and uses `printf`, `scanf`, and `for` loops to check if a number is prime. On the right, a terminal window shows the program's output for the input 3, confirming it is a prime number and displaying the process return status and execution time.

```
#include <stdio.h>
int main()
{
    int n,i,count=0;
    printf("Enter a number: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            count++;
        }
    }
    if(count==2)
    printf("%d is a prime Number\n",n);
    else
    printf("%d is Not a prime Number\n",n);
    return 0;
}
```

Enter a number: 3  
3 is a prime Number

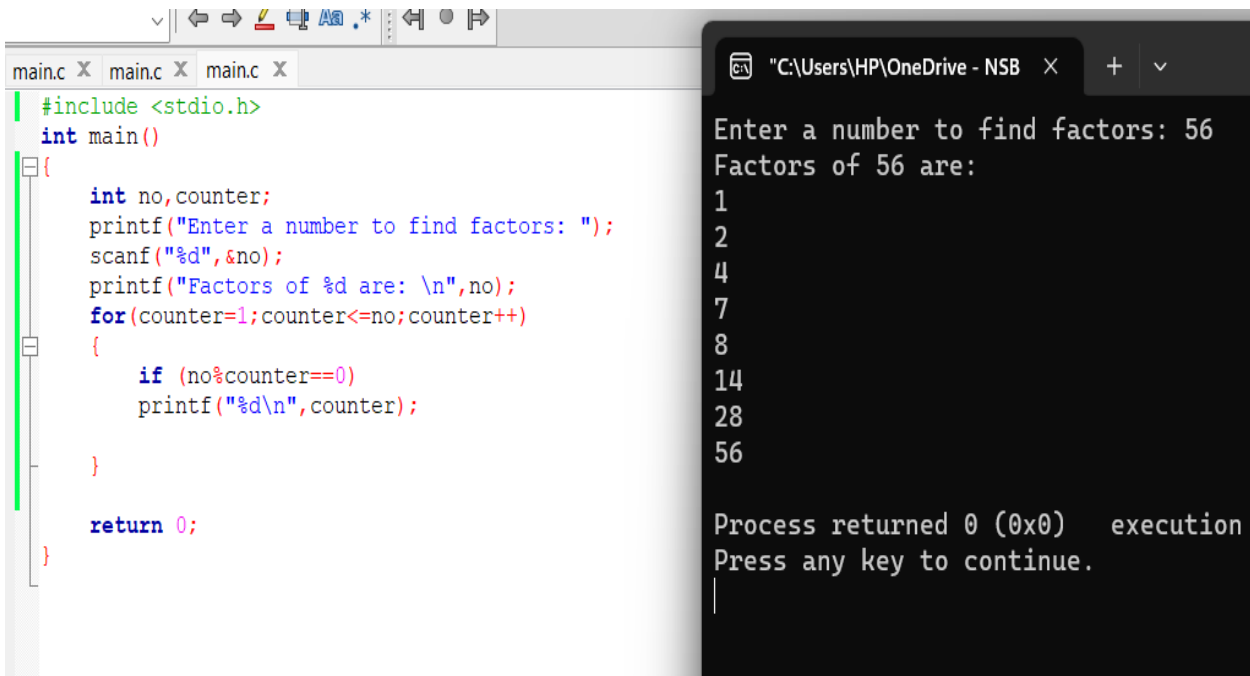
Process returned 0 (0x0) ex  
1.678 s  
Press any key to continue.

Q12)

```
#include <stdio.h>
int main()
{
    int no,counter;
    printf("Enter a number to find factors: ");
    scanf("%d",&no);
    printf("Factors of %d are: \n",no);
    for(counter=1;counter<=no;counter++)
    {
        if (no%counter==0)
            printf("%d\n",counter);

    }

    return 0;
}
```



The screenshot displays a C program in a text editor and its execution output in a terminal window. The code in the editor is as follows:

```
#include <stdio.h>
int main()
{
    int no,counter;
    printf("Enter a number to find factors: ");
    scanf("%d",&no);
    printf("Factors of %d are: \n",no);
    for(counter=1;counter<=no;counter++)
    {
        if (no%counter==0)
            printf("%d\n",counter);

    }

    return 0;
}
```

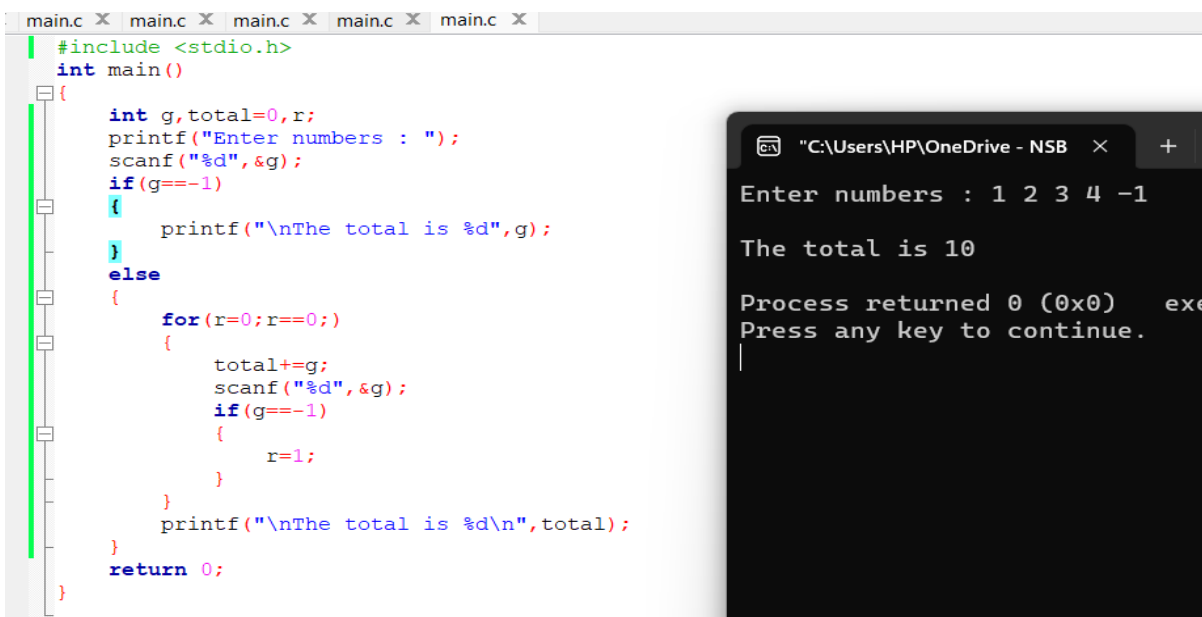
The terminal window shows the program's execution with the input 56. The output lists the factors of 56: 1, 2, 4, 7, 8, 14, 28, and 56. The process then returns 0, indicating successful execution.

```
"C:\Users\HP\OneDrive - NSB" x + v
Enter a number to find factors: 56
Factors of 56 are:
1
2
4
7
8
14
28
56

Process returned 0 (0x0)   execution
Press any key to continue.
|
```

Q12)

```
#include <stdio.h>
int main()
{
    int g,total=0,r;
    printf("Enter numbers : ");
    scanf("%d",&g);
    if(g== -1)
    {
        printf("\nThe total is %d",g);
    }
    else
    {
        for(r=0;r==0;)
        {
            total+=g;
            scanf("%d",&g);
            if(g== -1)
            {
                r=1;
            }
        }
        printf("\nThe total is %d\n",total);
    }
    return 0;
}
```



The screenshot displays a C program and its execution. On the left, a text editor window shows the source code for 'main.c'. The code defines a function 'main' that prompts the user to enter numbers. It uses a loop to repeatedly add the input numbers to a total until the user enters -1. On the right, a terminal window shows the program's output. The user has entered the numbers 1, 2, 3, 4, and -1. The program correctly calculates the total as 10 and displays it. The terminal also shows the standard Windows command prompt messages: 'Process returned 0 (0x0)' and 'Press any key to continue.'

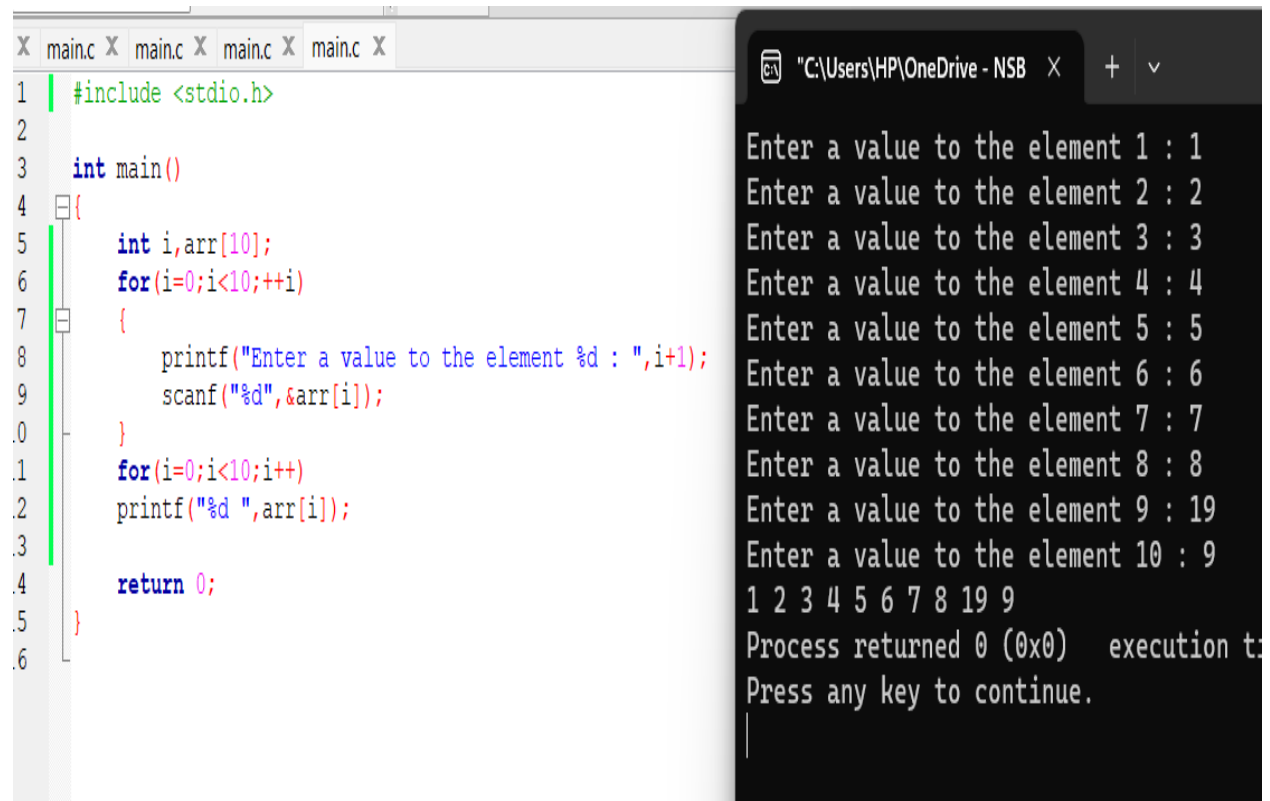
```
main.c X main.c X main.c X main.c X main.c X
#include <stdio.h>
int main()
{
    int g,total=0,r;
    printf("Enter numbers : ");
    scanf("%d",&g);
    if(g== -1)
    {
        printf("\nThe total is %d",g);
    }
    else
    {
        for(r=0;r==0;)
        {
            total+=g;
            scanf("%d",&g);
            if(g== -1)
            {
                r=1;
            }
        }
        printf("\nThe total is %d\n",total);
    }
    return 0;
}
```

```
"C:\Users\HP\OneDrive - NSB" X +
Enter numbers : 1 2 3 4 -1
The total is 10
Process returned 0 (0x0)
Press any key to continue.
```

Q13)

```
#include <stdio.h>
int main()
{
    int i,arr[10];
    for(i=0;i<10;++i)
    {
        printf("Enter a value to the element %d : ",i+1);
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
        printf("%d ",arr[i]);

    return 0;
}
```



The screenshot shows a C program being executed in a Windows environment. The code is displayed in a text editor with line numbers 1 through 6. The program prompts the user to enter values for 10 elements of an array. The output shows the entered values: 1, 2, 3, 4, 5, 6, 7, 8, 19, and 9. The process returns 0 (0x0) and execution terminates. The user is prompted to press any key to continue.

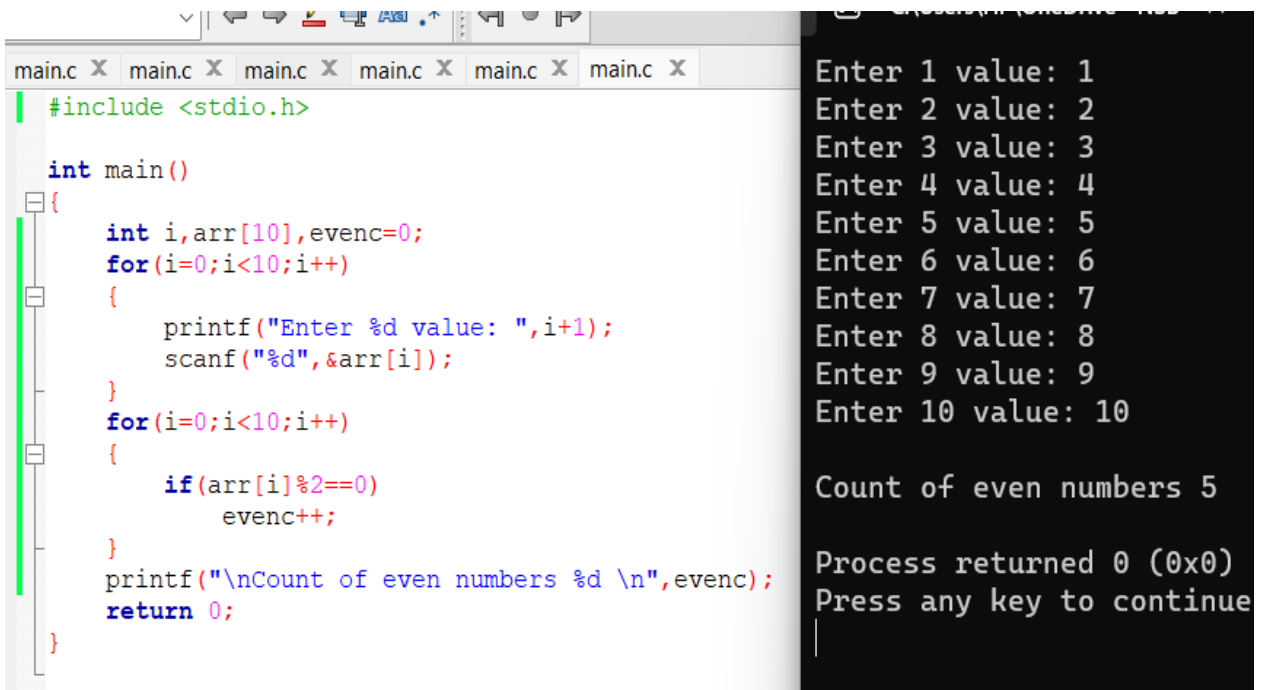
```
1 #include <stdio.h>
2
3 int main()
4 {
5     int i,arr[10];
6     for(i=0;i<10;++i)
7     {
8         printf("Enter a value to the element %d : ",i+1);
9         scanf("%d",&arr[i]);
10    }
11    for(i=0;i<10;i++)
12        printf("%d ",arr[i]);
13
14    return 0;
15 }
```

Enter a value to the element 1 : 1  
Enter a value to the element 2 : 2  
Enter a value to the element 3 : 3  
Enter a value to the element 4 : 4  
Enter a value to the element 5 : 5  
Enter a value to the element 6 : 6  
Enter a value to the element 7 : 7  
Enter a value to the element 8 : 8  
Enter a value to the element 9 : 19  
Enter a value to the element 10 : 9  
1 2 3 4 5 6 7 8 19 9  
Process returned 0 (0x0) execution terminated.  
Press any key to continue.



Q14)

```
#include <stdio.h>
int main()
{
    int i,arr[10],evenc=0;
    for(i=0;i<10;i++)
    {
        printf("Enter %d value: ",i+1);
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
    {
        if(arr[i]%2==0)
            evenc++;
    }
    printf("\nCount of even numbers %d \n",evenc);
    return 0;
}
```



The screenshot shows a C program running in a terminal window. The program prompts the user to enter 10 values. The user enters the numbers 1 through 10. The program then outputs the count of even numbers, which is 5. The terminal window also shows the process returning 0 (0x0) and a prompt to press any key to continue.

```
main.c X main.c X main.c X main.c X main.c X main.c X
#include <stdio.h>

int main()
{
    int i,arr[10],evenc=0;
    for(i=0;i<10;i++)
    {
        printf("Enter %d value: ",i+1);
        scanf("%d",&arr[i]);
    }
    for(i=0;i<10;i++)
    {
        if(arr[i]%2==0)
            evenc++;
    }
    printf("\nCount of even numbers %d \n",evenc);
    return 0;
}
```

```
Enter 1 value: 1
Enter 2 value: 2
Enter 3 value: 3
Enter 4 value: 4
Enter 5 value: 5
Enter 6 value: 6
Enter 7 value: 7
Enter 8 value: 8
Enter 9 value: 9
Enter 10 value: 10

Count of even numbers 5

Process returned 0 (0x0)
Press any key to continue
```

## Section B

Q1)

```
#include <stdio.h>
int main()
{
    {int no,counter=0,negative=0,positive=0,zero=0;

    for(counter;counter<10;counter++)
    {
        printf("Enter number : ");
        scanf("%d",&no);
        if(no==0)
        {
            zero++;
        }
        else if(no<0)
        {
            negative++;
        }
        else
        {
            positive++;
        }
    }
    printf("There are \n %d positive numbers \n %d negative number
        \n %d Zeros",positive,negative,zero);
    return 0;
}
}
```

```
main.c X main.c X main.c X main.c X main.c X main.c X main.c X
#include <stdio.h>
int main()
{
    {int no, counter=0, negative=0, positive=0, zero=0;

    for(counter; counter<10; counter++)
    {
        printf("Enter number : ");
        scanf("%d", &no);
        if(no==0)
        {
            zero++;
        }
        else if(no<0)
        {
            negative++;
        }
        else
        {
            positive++;
        }
    }
    printf("There are \n %d positive numbers \n %d negative number \n %d Zeros", positive, negative, zero);
    return 0;
}
```

"C:\Users\HP\OneDrive - NSB X + v

Enter number : 4  
Enter number : 5  
Enter number : 0  
Enter number : -34  
Enter number : 6  
Enter number : 7  
Enter number : 79  
Enter number : -9  
There are  
6 positive numbers  
3 negative number  
1 Zeros  
Process returned 0 (0x0) execution t  
s  
Press any key to continue.

Q2)

```
#include <stdio.h>
int main()
{
    int i,marks[10],max,min,total=0;
    float avg;
    printf("Enter the marks of 10 students\n\n");

    max=marks;
    min=marks;
    for(i=0;i<10;i++)
    {
        printf("Enter marks for student %d: ",i+1);
        scanf("%d",&marks[i]);
        total+=marks[i];
        if(i==0||marks[i]>max)
        {
            max=marks[i];
        }

        if(min>marks[i])
        {
            min=marks[i];
        }
    }
    avg=(float)total/10;

    printf("\nThe maximum number is %d\n",max);
    printf("The minimum number is %d\n",min);
    printf("The average is %2f\n",avg);
    return 0;
}
```

```

1  #include <stdio.h>
2  int main()
3  {
4      int i,marks[10],max,min,total=0;
5      float avg;
6      printf("Enter the marks of 10 students\n\n");
7
8      max=marks;
9      min=marks;
10     for(i=0;i<10;i++)
11     {
12         printf("Enter marks for student %d: ",i+1);
13         scanf("%d",&marks[i]);
14         total+=marks[i];
15         if(i==0||marks[i]>max)
16         {
17             max=marks[i];
18         }
19
20         if(min>marks[i])
21         {
22             min=marks[i];
23         }
24     }
25     avg=(float)total/10;
26
27     printf("\nThe maximum number is %d\n",max);
28     printf("The minimum number is %d\n",min);
29     printf("The average is %2f\n",avg);
30     return 0;
31 }

```

"C:\Users\HP\OneDrive - NSB" X + v

Enter the marks of 10 students

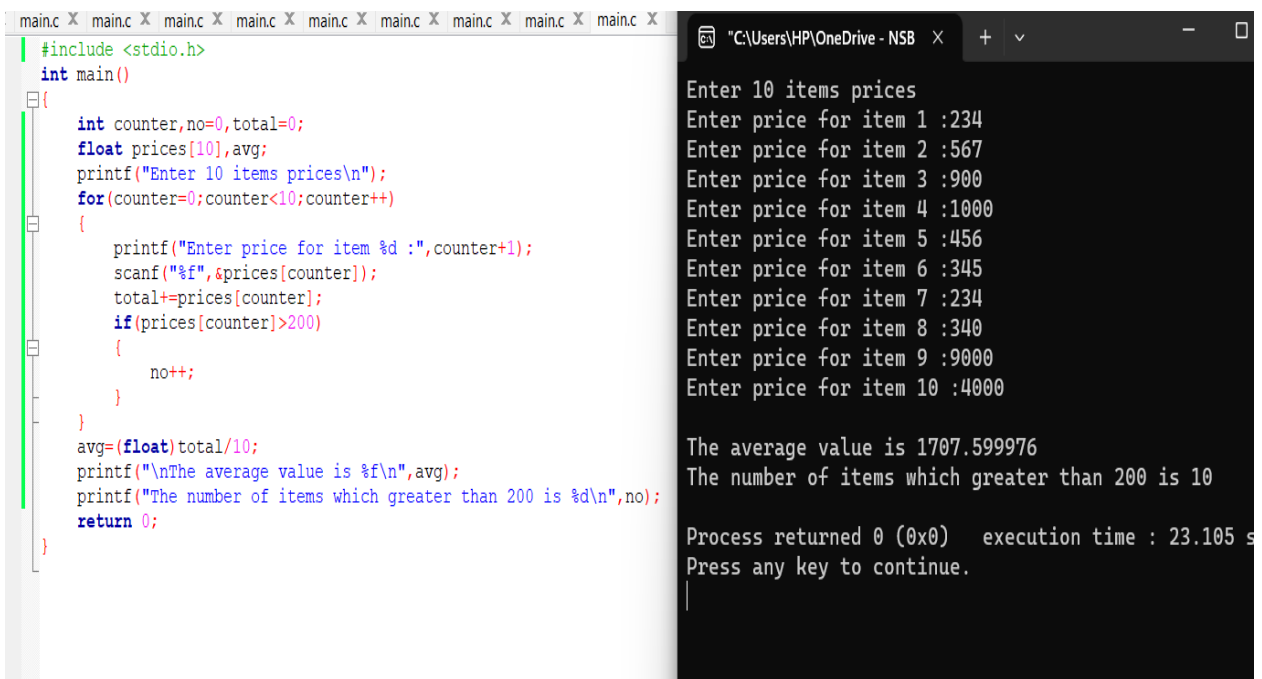
Enter marks for student 1: 34  
Enter marks for student 2: 56  
Enter marks for student 3: 78  
Enter marks for student 4: 90  
Enter marks for student 5: 23  
Enter marks for student 6: 22  
Enter marks for student 7: 66  
Enter marks for student 8: 56  
Enter marks for student 9: 45  
Enter marks for student 10: 12

The maximum number is 90  
The minimum number is 12  
The average is 48.200001

Process returned 0 (0x0) execution  
Press any key to continue.

Q3)

```
#include <stdio.h>
int main()
{
    int counter,no=0,total=0;
    float prices[10],avg;
    printf("Enter 10 items prices\n");
    for(counter=0;counter<10;counter++)
    {
        printf("Enter price for item %d :",counter+1);
        scanf("%f",&prices[counter]);
        total+=prices[counter];
        if(prices[counter]>200)
        {
            no++;
        }
    }
    avg=(float)total/10;
    printf("\nThe average value is %f\n",avg);
    printf("The number of items which greater than 200 is %d\n",no);
    return 0;
}
```



The screenshot displays a C program being executed. On the left, a text editor shows the source code, which includes headers, variable declarations, a loop to input 10 prices, a conditional count of items exceeding 200, and calculations for the average and final output. On the right, a terminal window shows the program's execution. It prompts for 10 item prices, which are entered as 234, 567, 900, 1000, 456, 345, 234, 340, 9000, and 4000. The program then outputs the average value as 1707.599976 and the count of items greater than 200 as 10. The terminal also shows the process returning 0 and an execution time of 23.105 seconds.

```
main.c x main.c x main.c x main.c x main.c x main.c x main.c x main.c x main.c x
#include <stdio.h>
int main()
{
    int counter,no=0,total=0;
    float prices[10],avg;
    printf("Enter 10 items prices\n");
    for(counter=0;counter<10;counter++)
    {
        printf("Enter price for item %d :",counter+1);
        scanf("%f",&prices[counter]);
        total+=prices[counter];
        if(prices[counter]>200)
        {
            no++;
        }
    }
    avg=(float)total/10;
    printf("\nThe average value is %f\n",avg);
    printf("The number of items which greater than 200 is %d\n",no);
    return 0;
}

"C:\Users\HP\OneDrive - NSB x + v - □
Enter 10 items prices
Enter price for item 1 :234
Enter price for item 2 :567
Enter price for item 3 :900
Enter price for item 4 :1000
Enter price for item 5 :456
Enter price for item 6 :345
Enter price for item 7 :234
Enter price for item 8 :340
Enter price for item 9 :9000
Enter price for item 10 :4000

The average value is 1707.599976
The number of items which greater than 200 is 10

Process returned 0 (0x0)   execution time : 23.105 s
Press any key to continue.
|
```

Q4)

```
#include <stdio.h>
int main()
{
    int empno,counter=0;
    float basicsalary;
    printf("Enter the employee number and basicsalary(enter -999 to
stop employee number) \n");

    while(1)
    {   printf("\nEmployee no: ");
        scanf("%d",&empno);

        if(empno==-999)
        {
            break;
        }
        printf("Basic Salary: ");
        scanf("%f",&basicsalary);

        if(basicsalary>=5000)
        {
            counter++;
        }
    }
    printf("\nThe number of employees whose Basic Salary >=5000 is
        %d\n",counter);
    return 0;
}
```

```

1  #include <stdio.h>
2  int main()
3  {
4      int empno, counter=0;
5      float basicsalary;
6      printf("Enter the employee number and basicsalary(enter -999 to stop employee number) \n");
7
8      while(1)
9      { printf("\nEmployee no: ");
10         scanf("%d", &empno);
11
12         if(empno==-999)
13         {
14             break;
15         }
16         printf("Basic Salary: ");
17         scanf("%f", &basicsalary);
18
19         if(basicsalary>=5000)
20         {
21             counter++;
22         }
23     }
24     printf("\nThe number of employees whose Basic Salary >=5000 is %d\n", counter);
25     return 0;
26 }
27

```

```

"C:\Users\HP\OneDrive - N
+ v - □ X

Enter the employee number and basicsalary(enter
-999 to stop employee number)

Employee no: 1
Basic Salary: 20000

Employee no: 2
Basic Salary: 90000

Employee no: 3
Basic Salary: 50000

Employee no: -999

The number of employees whose Basic Salary >=50
00 is 3

Process returned 0 (0x0)   execution time : 16.
704 s

```



```

Q5) #include <stdio.h>
int main()
{
    int empno,workhours,overtimepayment,counter=0,totalemp=0;
    printf("Enter your employee number hours worked(enter-999 to
stop employee number)\n");

    while(1)
    {
        printf("\nEnter Employee No: ");
        scanf("%d",&empno);

        if(empno==-999)
        {
            break;
        }
        printf("Enter Hours Work: ");
        scanf("%d",&workhours);

        totalemp++;

        if(workhours<=40)
        {
            overtimepayment=0;
        }
        else
        {
            overtimepayment=workhours-40;
            overtimepayment=(overtimepayment*150)+((40-workhours)*150);
        }

        if(overtimepayment)
        printf("Employee No: ",empno);
        printf("overtimepayment: ",overtimepayment);

        if(overtimepayment>=4000)
        {
            counter++;
        }

        float percentage=((float)counter/totalemp)*100;
    }
}

```

```
        printf("Perccenntage of employees with overtime payment  
exceeding Rs4000 : %2f ",percentage);  
    }  
    return 0;  
}
```